

Information Technology
IT-47-04

The Effect of Screen Time on University Students' Sleep Quality

A Quantitative Study Using the Z-Test for the Difference of Two Means



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Background & GAP

- Smartphones, laptops, and tablets are now inseparable from university students' lives.
- The WHO reports a significant rise in digital screen use among young people over the past decade.
- Sleep quality is a critical factor for academic performance, physical health, and mental well-being.

Research on this topic for Indonesian university students is still limited.
This study aims to fill that gap.

Research Question & Hypotheses

Research Question

Is there a real difference in sleep quality between students who spend more than 6 hours a day on screens and those who spend 6 hours or less?

Two possibilities tested:

- H_0 : There is no difference between the two groups.
- H_1 : There is a difference between the two groups.

Related Work

Study	Key Finding
Carter et al. (2016)	Heavy screen users had twice the risk of poor sleep.
Chang et al. (2015)	Light from screens at night reduces the body's sleep hormone (melatonin).
Demirci et al. (2015)	University students who use phones heavily tend to sleep worse and feel more stressed.

Gap Research

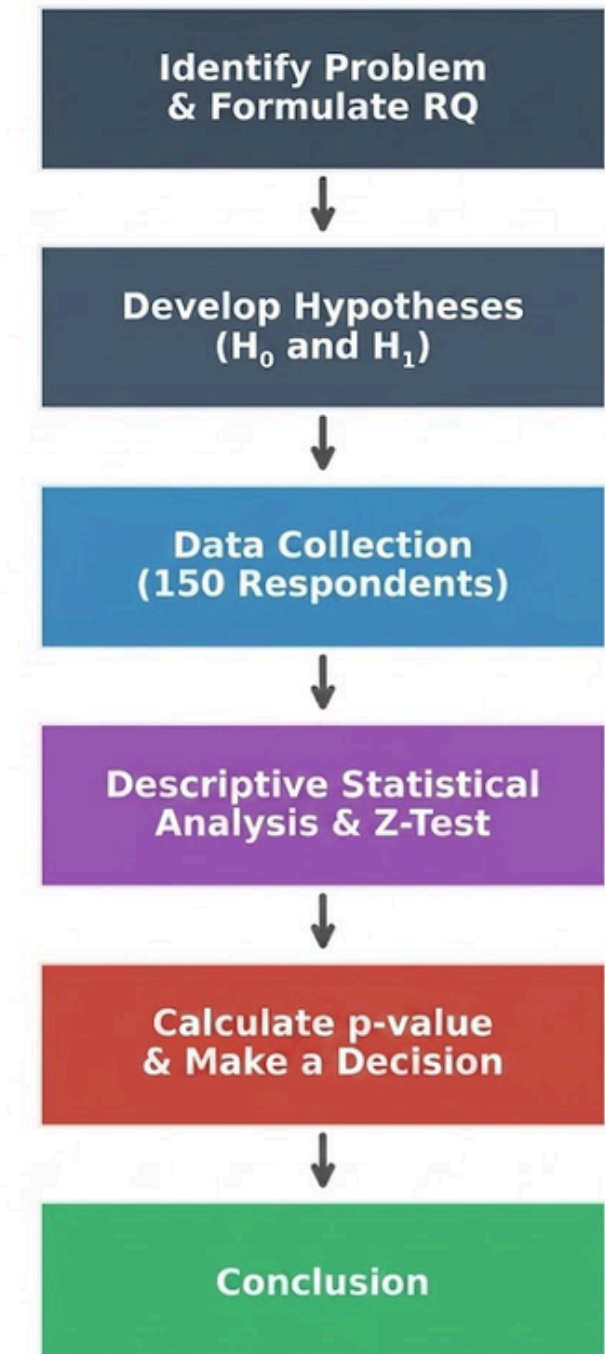
**Most of these studies were done in Western countries.
This study adds new data from Indonesian university students.**

Research Methodology Flow

About Data

150 university students filled out a questionnaire. They reported their daily screen time (in hours) and rated their sleep quality on a scale of 1 to 10.

Category	Range (hours/day)	n
Low	≤ 3	19
Medium	3 – 6	79
High	> 6	52



Results

Average Sleep Quality by Group (scale 1–10):

- Low screen time (≤ 3 hours): 7.39
- Medium screen time (3–6 hours): 6.28
- High screen time (> 6 hours): 4.68

Comparison of High vs. Non-High groups:

- High group average: 4.68
- Non-high group average: 6.49
- Difference: 1.82 points a substantial gap.

Pattern: The more screen time,
the lower the sleep quality.

Result of the test:

The chance this gap is just coincidence is essentially zero far below the 5% limit.

Higher screen time is linked to lower sleep quality.

Conclusion & Future Work

Main Findings:

- **Students with high screen time (more than 6 hours a day) had significantly lower sleep quality than other students.**
- **Screen time and sleep quality are strongly and negatively linked.**

Managing daily screen time is important for maintaining good sleep.

THANK YOU!

We would greatly appreciate any questions or discussion!